

Jeremy Hui

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EDUCATION

Rutgers University – Engineering Honors Academy

New Brunswick, NJ

B.S. Double Major in Electrical/Computer Engineering and Computer Science, Minor in Math Sep. 2022 – May 2026

Honors: Tau Beta Pi, Sophia Sheng Endowed Scholarship, Dean's List, The Dean's Scholarship

Relevant Courses: Hardware & Systems Security, Computer Architecture & Assembly Language, Operating Systems, Network-Centric Programming, Complexity Theory, Linear Systems & Signals, Electronic Devices

EXPERIENCE

HWSEL - Hardware Security Research

Sept. 2024 – Present

Research Assistant

New Brunswick, NJ

- Securing disaggregated memory through leveraging hardware-level isolation on Arm CCA processors.
- Training an HPC secure agent to automate setting 30+ BIOS parameters in confidential containers for DNNs.

Princeton-Intel Research Experience for Undergraduates

Jun. 2024 – Jul. 2024

Summer Research Intern

Princeton, NJ

- Investigated secure memory through testing Rowhammer-like attacks on OpenSSD 3D NAND Flash Memory.
- Ran experiments using C/microcode to find reverse mapping between aggressor and victim pages, and demonstrated the ability of directing 50,000 bit flips to specific cells per 1 million repeated write cycles.
- Experimented with and documented how BCH error correction codes on NAND Flash Memory can be bypassed.

Wireless Information Network Laboratory (WINLAB)

Jun. 2023 – Aug. 2023

Summer Research Intern

North Brunswick, NJ

- Implemented an experimentation platform about human interactivity with indoor spaces using Python and Linux, and effectively communicated results in 5-minute weekly presentations.
- Set up 25 Raspberry Pis to securely communicate environment data to be processed in Flask database by an LLM.
- Implemented Precision Time Protocol (PTP) to synchronize clocks on a nanosecond scale.

Rutgers University Course Staff

Sep. 2023 – Present

Head TA for Intro to CS, Learning Assistant for Intro to Computers for Engineers

New Brunswick, NJ

- Assisted over 600 students through holding office hours and managing the course website.
- Overlook operations for over 30 other course staff (learning assistants, graders, etc.).
- Led 2 group of 40+ students in learning MATLAB through weekly 80-minute active-learning recitations.

PROJECTS

Concurrent Web Proxy Server | C

Feb. 2025 – Apr. 2025

- Developed a web proxy that forwards requests between a browser and server using sockets, with two implementations: a multithread and a multi-process implementation for each new connection request.
- Created logging features for the proxy, with synchronization primitives and thread safety to prevent corruption.
- Implemented HTTPS tunneling to allow for CONNECT requests to relay data without interpreting encrypted data, and ensured proper termination when a connection closes or the tunnel is idle.

PC Composer | Node.js, SQLite3, React, AWS S3, OpenAI, Express.js

Sept. 2024 – Dec. 2024

- Designed and managed a full stack web application using a React.js frontend and SQLite3 backend database to generate/purchase a PC build using an AI chatbot from 100+ computer parts.
- Used AWS S3, applied design patterns, and enforced object-oriented system specifications according to business goals/use cases in order to effectively coordinate an 8-member team. Applied an Agile development methodology.

Intro CS/Data Structures Assignment Developer | Java, Java Swing, Markdown

Jun. 2023 – Jul. 2024

- Developed 5 Java assignments 1000+ students promoting humanitarian applications of computer science.
- Created assignments write-ups that explained directions and 2 autograders to process student submissions.

TECHNICAL SKILLS

Languages: HTML, Javascript, Java, C/C++, MATLAB, Python, SQL (MySQL), RISC-V

Frameworks: React, Node.js, WordPress, Express.js, Jasmine

Tools: AWS EC2, gem5, Linux, GitHub, Docker, AWS S3, Vim, Vivado/Verilog, VS Code